BALINENI VISHNUVARDHAN

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Professional Summary

Computer Science undergraduate with hands-on experience in Generative AI, Machine Learning, and LLM-based application development. Experienced in building Retrieval-Augmented Generation (RAG) systems, AI safety guardrails, and intelligent applications using LangChain, LangGraph, and modern AI frameworks. Strong foundation in Data Structures and Algorithms with an interest in scalable AI systems, LLM engineering, and software development.

Education

Indian Institute of Information Technology(IIIT), Nuzvid — 2023–2027
B.Tech in Computer Science and Engineering , Andhra Pradesh

RGUKT ,Nuzvid — 2021–2023
Pre University Course(Intermediate) , Andhra Pradesh

Professional Experience

Full Stack Developer Intern — Present
Nabes Pvt Ltd – Remote

- Engineered scalable Human Resource Management System (HRMS) using React and Spring Boot microservices, reducing API response time by 30%
- Implemented Role-Based Access Control (RBAC) for 6 user roles with granular permissions and audit logging, ensuring enterprise-grade security compliance
- Optimized PostgreSQL database queries and implemented connection pooling, improving payroll processing performance by 40%
- Built automated payroll workflows reducing manual calculation errors by 90% and integrated multi-region compliance for India and US tax regulations

Technical Skills

Languages:	Java, Python, SQL
Generative AI:	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Workflows, AI Guardrails
Machine Learning:	Regression, Classification, Clustering, Model Training, Hyperparameter Tuning, Model Evaluation
Deep Learning:	Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), PyTorch, Keras
Data Processing:	Data Cleaning, Feature Engineering, EDA, Pandas, NumPy, Scikit-learn
Web Development:	React, Spring Boot, REST APIs, CSS
Databases:	MySQL, PostgreSQL, ChromaDB
Core CS:	Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, OOP
Tools:	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Vercel, Maven

Projects

Self-Healing RAG Pipeline — GitHub
Generative AI | LangGraph | RAG

- Built an advanced Retrieval-Augmented Generation (RAG) system with self-correction capabilities using LangGraph, enabling iterative retrieval and answer validation workflows
- Designed a critic agent to evaluate whether generated responses were grounded in retrieved document chunks, reducing hallucinated outputs through automated verification
- Implemented cyclical retry logic with query reformulation, allowing the system to re-retrieve context when answer quality failed validation instead of generating unsupported responses
- Developed a stateful multi-agent workflow using vector search, document chunking, embeddings, and conditional graph routing for reliable question answering over private datasets

***Technologies:** Python, LangChain, LangGraph, ChromaDB, Vector Embeddings, RAG, Prompt Engineering, LLMs*

LLM Guardrails Gateway GitHub

Generative AI | LLM Safety | Middleware

- Built a middleware layer for Large Language Models (LLMs) to enforce safety, compliance, and structured output validation before and after model inference
- Implemented input guardrails to detect prompt injection attacks, jailbreak attempts, and Personally Identifiable Information (PII) leakage using pattern matching and rule-based validation
- Designed output validation pipelines ensuring schema-compliant JSON responses, toxicity filtering, and topic relevance checks with automatic retry mechanisms
- Developed a configurable YAML-based policy engine enabling non-technical users to define custom restrictions such as blocking medical advice, enforcing citations, and restricting competitor discussions

Technologies: *Python, FastAPI, LangChain, YAML, Pydantic, Prompt Engineering, LLMs, JSON Validation*

Multi-Model Emotion Detection GitHub

Deep Learning | CNN | NLP

- Built CNN-based emotion recognition system achieving 85% accuracy across 6 emotion classes using TensorFlow and Keras and feature engineering
- Processed 10,000+ training images with including face detection, augmentation (rotation/zoom/flip), and normalization techniques
- Implemented hyperparameter tuning and model optimization improving precision to 90%

Technologies: *Python, TensorFlow, Keras, CNN, NumPy,Pandas*

HRMS Tool (Corporate Project) GitHub

Enterprise Human Resource Management System

- Developed full-stack HRMS supporting 500+ employees with multi-currency payroll, region-specific compliance (India/US), and automated tax calculations
- Engineered RBAC system with 6 role hierarchies and audit trails; built responsive React dashboard with role-based component rendering
- Optimized backend microservices with 95% test coverage using JUnit and Mockito, implementing JWT authentication and secure session management

Technologies: *React, Spring Boot, MySQL, Hibernate, JWT, REST APIs, Maven*

Achievements & Certifications

- Solved 150+ DSA problems across arrays, stacks,queues,linkedlist, trees, and dynamic programming on LeetCode
- Secured 3rd Place in Potti Sriramulu College Hackathon among 50+ teams for developing a Multi-Disease Prediction System using Machine Learning
- Completed NPTEL Certification in Artificial Intelligence, covering core concepts in intelligent systems and machine learning
- Completed NPTEL Certification in Cloud Computing, focusing on cloud architecture, virtualization, and distributed systems